

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Bearings, Scale Drawing & Con- structions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

BEARINGS, SCALE DRAWING & CONSTRUCTIONS · CH4 · L3



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The Map

This lesson collapses Bearings, Scale Drawing & Constructions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Draw North first

TRIGGER *"draw", "plot", "construct", a diagram/grid*

02 Measure clockwise bearing

TRIGGER *"bearing", "north", "scale drawing"*

03 Use scale before real distance

TRIGGER *"bearing", "north", "scale drawing"*

04 Construct the locus

TRIGGER *"draw", "plot", "construct", a diagram/grid*

05 Use bearings

TRIGGER *"bearing", "north", "scale drawing"*

06 Construction answer

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Draw North first

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q16

Use bearings

This is a reasoning step: write one clear sentence, then continue the calculation.

Resolve

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

Finish: 140° .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

16 The diagram shows the positions of three ships, A, B and C. A 150 m 275 m North North 120 degrees B C Diagram NOT accurately drawn Ship B is due north of ship A. The bearing of ship C from ship A is 120 degrees Calculate the bearing of ship C from ship B. Give your answer correct to the nearest degree. degrees

YOUR SOLUTION

02 Measure clockwise bearing

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

03 Use scale before real distance

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

1 cm represents 200 m

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q5

The distance from to is

This is a reasoning step: write one clear sentence, then continue the calculation.

The scale is

1 cm represents 200 m

Finish: Draw C 3.5 cm from A on a bearing of 110° , and the scale is 1 : 20000.

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

5 The scale diagram shows the position on a map of a house, A A North Scale: 1 cm represents 200 metres House C is on a bearing of 110 degrees from A The distance from A to C is 700 m (a) Mark the position of C on the diagram with a cross (times) Label your cross C (3) (b) Write the scale of the map in the form 1 : n 1 : (1)

YOUR SOLUTION

04 Construct the locus

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

05 Use bearings

TRIGGER *"bearing", "north", "scale drawing"*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q16

Use bearings

This is a reasoning step: write one clear sentence, then continue the calculation.

Resolve

$$\text{east component} = 275 \sin 120^\circ = 238.16 \dots$$

Finish: 140° .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

16 The diagram shows the positions of three ships, A, B and C. A 150 m 275 m North North 120 degrees B C Diagram NOT accurately drawn Ship B is due north of ship A. The bearing of ship C from ship A is 120 degrees Calculate the bearing of ship C from ship B. Give your answer correct to the nearest degree. degrees

YOUR SOLUTION

06 Construction answer

TRIGGER *"draw", "plot", "construct", a diagram/grid*

BECOMES A Bearings, Scale Drawing & Constructions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q5

Construction answer

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: The ray from A through the arc intersection is the angle bisector of BAC .

FORMULA BANK

Bearings are measured clockwise from North using three digits. Scale: drawing : real.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 Use ruler and compasses to construct the bisector of angle BAC. You must show all your construction lines. B A C

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"draw", "plot", "construct", a dia- gram/grid	Draw North first
"bearing", "north", "scale drawing"	Measure clockwise bearing
"bearing", "north", "scale drawing"	Use scale before real distance
"draw", "plot", "construct", a dia- gram/grid	Construct the locus
"bearing", "north", "scale drawing"	Use bearings
"draw", "plot", "construct", a dia- gram/grid	Construction answer

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.